

observation that occupies the $(n + 1)/2$ position. In an even-numbered sample, we define the median as the mean of the values of the observations occupying the $n/2$ and $(n + 2)/2$ positions (the two middle observations).

Whether we use the calculation for an even- or odd-numbered sample, an equal number of observations lie above and below the median. A distribution has only one median. A potential advantage of the median is that, unlike the mean, outliers do not affect it.

The median, however, does not use all the information about the size of the observations; it focuses only on the relative position of the ranked observations. Calculating the median may also be more complex. Mathematicians express this disadvantage by saying that the median is less mathematically tractable than the mean.

The Mode

A third important measure of central tendency is the mode.

Definition of Mode. The mode is the most frequently occurring value in a dataset. A dataset can have more than one mode, or even no mode. When a dataset has a single value that is observed most frequently, its distribution is said to be **unimodal**. If a dataset has two most frequently occurring values, then it has two modes and its distribution is referred to as **bimodal**. When all the values in a dataset are different, the distribution has no mode because no value occurs more frequently than any other value.

Stock return data and other data from continuous distributions may not have a modal outcome. Exhibit 2 presents the frequency distribution of the daily returns for the EAA Equity Index over the past five years.

Exhibit 2: Frequency Distribution for Daily Returns of EAA Equity Index

Return Bin (%)	Absolute Frequency	Relative Frequency (%)	Cumulative Absolute Frequency	Cumulative Relative Frequency (%)
-5.0 to -4.0	1	0.08	1	0.08
-4.0 to -3.0	7	0.56	8	0.64
-3.0 to -2.0	23	1.83	31	2.46
-2.0 to -1.0	77	6.12	108	8.59
-1.0 to 0.0	470	37.36	578	45.95
0.0 to 1.0	555	44.12	1,133	90.06
1.0 to 2.0	110	8.74	1,243	98.81
2.0 to 3.0	13	1.03	1,256	99.84
3.0 to 4.0	1	0.08	1,257	99.92
4.0 to 5.0	1	0.08	1,258	100.00

A histogram for the frequency distribution of these daily returns was shown in Exhibit 1. The modal interval always has the highest bar in the histogram; in this case, the modal interval is 0.0 to 0.9 percent, and this interval has 493 observations out of a total of 1,258 observations.